

GRADUATION PROJECT PROPOSAL

PROJECT NAME	Network Automation And Security	
PROJECT Supervisor	DR / Mahaba	
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SUMMARY	• This project aims to design and implement a robust network automation and security solution using a topology that is two branches includes routers, switches, firewalls, and other security measures. • The objective is to enhance network efficiency while ensuring comprehensive security against potential threats and to configure your network more easily by replacing the command line mode with a GUI application.
PROBLEM STATEMENT	• Its so difficult to configure each device in your network by using command line especially if it is a large networks so we need the Automated networks , Also , as organizations increasingly rely on digital infrastructure, network vulnerabilities have become a significant concern. Traditional manual network management is prone to human error and inefficiencies, leading to security breaches and downtime. This project addresses the need for an automated, secure network system that can adapt to evolving threats.
EXPECTED CHALLENGES	1. Integration of Diverse Technologies: Ensuring seamless communication between different network devices and security solutions. 2. Scalability: Designing a solution that can grow with organizational needs without compromising security. 3. Security Threats: Identifying and mitigating emerging cyber threats during and post-implementation. 4. User Training: Ensuring that staff are adequately trained to operate and manage the new system.

PROPOSED SOLUTION	We propose the implementation of a network automation and security framework that incorporates: • Automated Configuration Management: Using netmiko library in python to manage devices configurations . • Real-time Security Monitoring: Implementing SIEM (Security Information and Event Management) systems to analyze security data. • Intrusion Detection Systems (IDS): Using tools like Snort to detect and respond to threats. • Firewall Management: Deploying next-gen firewalls to filter traffic and prevent unauthorized access.
COMPETITORS/MAIN WORK DONE IN THE LITERATURE	Network automated by using python with svi,ospf,monitor,vlans and router on stack
DESIRED OUTCOME	The project aims to deliver a fully automated network with its security that: • Reduces manual configuration errors in CLI. • Enhances security protocols. • Provides detailed logging and reporting for compliance purposes. • Improves network performance and uptime. • Mitigating network attacks. • Make the configuring operations more easier.
WHAT DO YOU NEED TO LEARN IF ANY	• Network infrastructure (Knowledge of TCP/IP and basic network Configurations). • Network Automation Tools: Proficiency in Netmiko Paramiko libraries . • Security Protocols: Understanding of firewalls, IDS/IPS, and SIEM systems. • Scripting Languages: Skills in Python for automation tasks. GUI with Python.
EXPECTED TOOLS/TECHNOLOGIES NEEDED	• Virtualization programs like: we will use EVE , VMWARE . • A code editor : we will use VSCode to run python . • Python Libraries : ▪ for GUI : Tkinter and customtkinter libraries. ▪ for Automation : Netmiko and Paramiko libraries.

PROJECT SCHEDULE/TIME PLAN	Phase	Duration
	Research and Literature Review	2 weeks
	Design Network Topology	1 week
	Implementation	6 weeks
	Testing and Evaluation	2 weeks
	Documentation and Reporting	1 week
PROJECTED TEAM AND RESOURCE REQUIREMENTS IF ANY	• Team Members: 5-6 students with backgrounds in networking, security, and programming. • Resources : Access to network with virtualization, testing environments, and software licenses if required.	